Modèles stochastiques en finance
Séance 7
Modèles à sauts.

Ioane Muni Toke

Laboratoire MICS, Equipe FiQuant, CentraleSupélec, France

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Programme

Poisson processes

A brief introduction to stochastic calculus with jumps

Adding jumps to the Black & Scholes model ($\longrightarrow$ TP Question VI)

Adding jumps to the stochastic volatility models

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Poisson process : Definition

- Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space.
- Let $(\tau_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d random variables, with respect to an exponential distribution with parameter $\lambda \in \mathbb{R}_+^*$.
- Let us set $S_n = \sum_{i=1}^n \tau_i$, $n \in \mathbb{N}^*$ and

$$N_t = \sum_{n\in\mathbb{N}^*} \mathbf{1}_{S_n \leq t}, \quad t \in \mathbb{R}_+,$$

  or equivalently

$$N_t = \max\{n \in \mathbb{N} : S_n \leq t\}, \quad t \in \mathbb{R}_+.$$

- $(N_t)_{t\geq 0}$ is called a (time-homogeneous) *Poisson process* with intensity λ. It is a *counting process*.

Poisson processes : Basic properties I

Proposition 7.1 (Basic properties)

Let $(\tau_i)_{i\in\mathbb{N}}$, $(S_n)_{n\in\mathbb{N}^*}$ *and* $(N_t)_{t\geq 0}$ *defined as above.*

(a) *For any* $n \in \mathbb{N}^*$, S_n *is Gamma distributed with shape parameter* n *and rate parameter* λ, *i.e. with density* $x \mapsto \frac{(\lambda x)^{n-1}}{(n-1)!}\lambda e^{-\lambda x}\mathbf{1}_{\mathbb{R}_+}(x)$.

(b) *For any* $t \in \mathbb{R}_+$, N_t *is Poisson distributed with parameter* λt, *i.e.* $\mathbf{P}(N_t = k) = \frac{(\lambda t)^k}{k!}e^{-\lambda t}$, $k \in \mathbb{N}$.

(c) *Increments of the Poisson process* $(N_t)_{t\geq 0}$ *are independent, i.e. for all* $n \in \mathbb{N}^*$ *and all strictly increasing finite sequence* $t_0 < t_1 < \ldots < t_n$, *the random variables* $N_{t_1}, N_{t_2} - N_{t_1}, \ldots, N_{t_n} - N_{t_{n-1}}$ *are mutually independent.*

Poisson processes : Basic properties II

Proposition 7.2 (Basic properties, cont'd)

Let $(\tau_i)_{i\in\mathbb{N}}$, $(S_n)_{n\in\mathbb{N}^*}$ *and* $(N_t)_{t\geq 0}$ *defined as above.*

(a) *Increments of the Poisson process* $(N_t)_{t\geq 0}$ *are stationary, i.e. for any* $s < t$, $N_t - N_s$ *and* N_{t-s} *have the same distribution, which is Poisson with parameter* $\lambda(t-s)$.

(b) *For any* $t \geq 0$, *the characteristic function of the random variable* N_t *is*

$$\varphi_{N_t}(u) = \mathbf{E}\left[e^{iuN_t}\right] = \exp\left\{\lambda t(e^{iu} - 1)\right\}.$$

(c) *The stochastic process* $(M_t)_{t\geq 0}$ *defined by* $M_t = N_t - \lambda t$, $t \geq 0$ *is called the compensated Poisson process and is a martingale with respect to* $\mathbf{P}$.

Compound Poisson Process I

- Let $(N_t)_{t\geq 0}$ be a Poisson process and let $(Z_i)_{i\in\mathbb{N}^*}$ be a sequence of independent and identically distributed random variables.
- Let us set

$$Q_t = \sum_{i=1}^{N_t} Z_i, \quad t \geq 0.$$

  $(Q_t)_{t\in\mathbb{R}_+}$ is called a compound Poisson process.
- (This is not a counting process since jumps are not unit-sized.)

Compound Poisson Process II

Proposition 7.3 (Compound Poisson process)

(a) *Increments of* $(Q_t)_{t\in\mathbb{R}_+}$ *are independent and stationary.*

(b) *For any* $t \geq 0$, *the characteristic function of the random variable* Q_t *is*

$$\varphi_{Q_t}(u) = E\left[e^{iuQ_t}\right] = \exp\left\{\lambda t(\varphi_Z(u) - 1)\right\},$$

where φ_Z *is the characteristic function of the (common) distribution of the* Z_i*'s.*

(c) *For all* $t \geq 0$, $\mathbf{E}[Q_t] = \mathbf{E}[N_t]\mathbf{E}[Z_1](= \lambda t\mathbf{E}[Z_1])$ *and* $\mathbf{Var}[Q_t] = \mathbf{E}[N_t]\mathbf{V}[Z_1](= \lambda t\mathbf{V}[Z_1])$.

(d) *The compensated compound Poisson process* $(R_t)_{t\in\mathbb{R}_+}$ *defined by* $R_t = Q_t - \lambda\mathbf{E}[Z_1]t$ *is a martingale with respect to* $\mathbf{P}$.

Compound Poisson Process III

- We can easily construct a compound Poisson process with a finite number of jump sizes by summing standard Poisson processes.

Proposition 7.4 (Compound Poisson process with a finite number of jump sizes)

Let $N_t^1, \ldots, N_t^M$ be M independent time-homogeneous Poisson processes with intensity $\lambda_m > 0$, $m = 1, \ldots, M$. Let $-1 < y_1 < y_2 < \ldots < y_M$ be M non-zero real numbers. If we set $N_t = \sum_{m=1}^M N_t^m$ and $Q_t = \sum_{m=1}^M y_m N_t^m$, then:

(a) *(N_t) is a Poisson process with intensity $\lambda = \sum_{m=1}^M \lambda_m$;*

(b) *(Q_t) is a compound Poisson process equal in distribution to $\sum_{i=1}^{N_t} Y_i$, where (Y_i) is an i.i.d. sequence of random variables such that $\mathbf{P}(Y_i = y_m) = \frac{\lambda_m}{\lambda}$ for all $m = 1, \ldots, M$.*

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A stochastic integral for jump processes I

- Let us now consider a pure jump process $(J_t)_{t\in[0,T]}$ with finite activity. Let us set $\Delta J_t = J_t - J_{t-}$. Obviously ΔJ_t is zero if there is no jump at time t, and equal to the size of the jump otherwise.
- Let $(\theta_t)_{t\in[0,T]}$ be a simple left-continuous process : $\theta_t = \sum_{i=1}^n \alpha_i \mathbf{1}_{]t_{i-1},t_i]}(t)$ where $0 = t_0 < t_1 < \ldots < t_n = T$ is a partition of $[0, T]$ et $\forall i, \alpha_i$ is a square-integrable $\mathcal{F}_{t_{i-1}}$-measurable random variable.
- A natural way to define the stochastic integral of the simple process $(\theta_t)_{t\in[0,T]}$ with respect to $(J_t)_{t\in[0,T]}$ is to set

$$\int_0^t \theta_s \, dJ_s = \sum_{i=1}^n \alpha_i \left(J_{t_i} - J_{t_{i-1}}\right).$$

A stochastic integral for jump processes II

- Now let $(\varphi_t)_{t\in[0,T]}$ be a *left-continuous* adapted process. $(\varphi_t)_{t\in[0,T]}$ is a limit of simple processes $(\varphi_t^{(n)})_{t\in[0,T]}$, $n \geq 0$, that can be integrated as

$$\int_0^t \varphi_s^{(n)}\, dJ_s = \sum_{i=1}^{n} \alpha_i^{(n)} \left(J_{t_i^{(n)}} - J_{t_{i-1}^{(n)}} \right).$$

- Since J has a finite activity, it is quite intuitive that as $n \to +\infty$,

$$\int_0^t \varphi_s^{(n)}\, dJ_s \to \sum_{0<s\leq t} \varphi_s (J_s - J_{s-}) = \sum_{0<s\leq t} \varphi_s \Delta J_s.$$

- We can therefore use as definition of a stochastic integral with respect to a pure jump process

$$\int_0^t \varphi_s\, dJ_s = \sum_{0<s\leq t} \varphi_s \Delta J_s.$$

Processes with jumps I

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be probability space with some filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$. Let B be a Brownian motion with respect to the filtration $(\mathcal{F}_t)_{0 \leq t \leq T}$ and the probability $\mathbf{P}$. From now on, we consider the following class of processes:

Definition 7.5 (Processes with jumps)

$(X_t)_{t \in [0,T]}$ *is a process with jumps if it is written*

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dB_s + J_t,$$

where μ et σ are $\mathcal{F}_t$-adapted stochastic processes, and $(J_t)_{t \in [0,T]}$ is a right-continuous, $\mathcal{F}_t$-adapted, purely discontinuous process with finite activity.

- In our financial model below, (J_t) will be a compound Poisson process.

Processes with jumps II

- Let $(\varphi_t)_{t\in[0,T]}$ be a *left-continuous* adapted process. The stochastic integral of $(\varphi_t)_{t\in[0,T]}$ with respect to the process with jumps X is

$$\int_0^t \varphi_s dX_s = \int_0^t \varphi_s \mu_s ds + \int_0^t \varphi_s \sigma_s dB_s + \sum_{0<s\leq t} \varphi_s \Delta J_s.$$

$\left(\int_0^t \varphi_s dX_s\right)$ is right-continuous.

Theorem 7.6 (Martingale property of stochastic integrals)

If the process with jumps X is a martingale, then for every $\mathcal{F}_t$-adapted, left-continuous, stochastic process φ such that $\forall t \in [0, T]$, $\mathbb{E}\left[\int_0^t \sigma_s^2 \varphi_s^2 ds\right] < +\infty$, then the process $\left(\int_0^t \varphi_s dX_s\right)_t$ is an $\mathcal{F}_t$-martingale.

Quadratic variation of a process with jumps I

- For a given process $(X_t)_{t\in[0,T]}$, let $X_t^c = \int_0^t \mu_s ds + \int_0^t \sigma_s dB_s$ be its continuous part.

Proposition 7.7 (Quadratic variation of a process with jumps)

The quadratic variation of a process with jumps $(X_t)_{t\in[0,T]}$ is:

$$[X]_t = [X^c]_t + \sum_{0<s\leq t} (\Delta X(s))^2 = \int_0^t \sigma_s^2 \, ds + \sum_{0<s\leq t} (\Delta J_s)^2$$

- Similarly, the cross-variation of two processes with jumps X_1 and X_2 is written

$$[X_1, X_2]_t = [X_1^c, X_2^c]_t + \sum_{0<s\leq t} \Delta X_1(s)\Delta X_2(s).$$

Quadratic variation of a process with jumps II

- A Brownian motion and a *pure* jump process have thus a zero cross-variation.
- The cross-variation of two *pure* jump processes with *distinct* jumps is also zero.
- Let N be a Poisson process with intensity $\lambda > 0$ and set $M_t = N_t - \lambda t$. Then $[M]_t = [N]_t = N_t$ but $\langle M \rangle_t = \lambda t$.

An Itō formula with jumps I

We may now write the following Itō formula for our class of processes with jumps.

Theorem 7.8 (An Itō formula with jumps)

Let X be a stochastic process with jumps. Let $f : [0, T] \times \mathbb{R}^d \to \mathbb{R}$ be a function continuously differentiable in the first variable, and twice continuously differentiable in the following d variables. Then:

$$f(t, X_t) = f(0, X_0) + \int_0^t \frac{\partial f}{\partial t}(s, X_s)\, ds + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(s, X_s)\, dX_s^{i,c}$$
$$+ \sum_{i=1}^d \sum_{j=1}^d \frac{1}{2} \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(s, X_s)\, d\left[X^{i,c}, X^{j,c}\right]_s + \sum_{0<s\leq t} \left(f(s, X_s) - f(s, X_{s-})\right).$$

An Itō formula with jumps II

- Application 1 : Let $(\mathcal{F}_t)_{t\in\mathbb{R}_+}$ be the natural filtration of the Poisson process $(N_t)_{t\in\mathbb{R}_+}$. If $(W_t)_{t\in\mathbb{R}_+}$ is a $\mathcal{F}_t$-Brownian motion, then W and N are independent.
- Application 2 : an "integration-by-parts" formula for processes with jumps:

Theorem 7.9 (Integration by parts for processes with jumps)

Let X and Y be two processes with jumps. Then

$$\begin{aligned} X_t Y_t &= X_0 Y_0 + \int_0^t X_s \, dY_s^c + \int_0^t Y_s \, dX_s^c + [X^c, Y^c]_t + \sum_{0<s\leq t} (X_s Y_s - X_{s-} Y_{s-}) \\ &= X_0 Y_0 + \int_0^t X_{s-} \, dY_s + \int_0^t Y_{s-} \, dX_s + [X, Y]_t \end{aligned}$$

Effects of a change of measure on a Poisson process I

The Girsanov theorem relies upon the process

$$Z_t = \exp\left(-\int_0^t \theta_s \, dBs - \frac{1}{2}\int_0^t \theta_s^2 \, ds\right) = \exp\left(X_t^c - \frac{1}{2}\left[X^c\right]_t\right).$$

(Z_t) is known as the Doleans-Dade exponential of the continuous process $X_t^c = -\int_0^t \theta_s \, dBs$.

Proposition 7.10 (Doleans-Dade exponential of a process with jumps)

Let X be a process with jumps. Let us define the process

$$Z_t = \exp\left(X_t^c - \frac{1}{2}\left[X^c\right]_t\right) \prod_{0<s\leq t} (1 + \Delta X_s).$$

Then Z satisfies $dZ_t = Z_{t-} dX_t$, with $Z_0 = 1$.

Effects of a change of measure on a Poisson process II

Theorem 7.11 (Change of measure for Poisson processes)

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space equipped with a filtration $(\mathcal{F}_t)_{t\in[0,T]}$. Let N_t be a time-homogeneous Poisson process with intensity $\lambda^{\mathbf{P}} > 0$ with respect to the measure $\mathbf{P}$ and the filtration $(\mathcal{F}_t)_{t\in[0,T]}$. Let $\lambda^{\mathbf{Q}} \in \mathbb{R}_+^*$. Let us define the stochastic process Z by:

$$Z_t = e^{(\lambda^{\mathbf{P}} - \lambda^{\mathbf{Q}})t} \left(\frac{\lambda^{\mathbf{Q}}}{\lambda^{\mathbf{P}}} \right)^{N_t}.$$

Then Z is a $\mathcal{F}_t$-martingale with respect to the measure $\mathbf{P}$, satisfying $\mathbf{E}^{\mathbf{P}}[Z_t] = 1$. Furthermore, if $\mathbf{Q}$ is the measure defined with $Z_T = \frac{d\mathbf{Q}}{d\mathbf{P}}$, then (N_t) is a time-homogeneous Poisson process with intensity $\lambda^{\mathbf{Q}}$ with respect to the measure $\mathbf{Q}$.

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Adding compound Poisson jumps to the Black & Scholes model I

- We add jumps to the Black & Scholes model using a compound Poisson process with finitely-many jump sizes (see above Proposition 7.4). More precisely, let $N_t^1, \ldots, N_t^M$ be M independent time-homogeneous Poisson processes with intensity $\lambda_m > 0$ and let $-1 < y_1 < y_2 < \ldots < y_M$ be M non-zero real numbers.
- Let $N_t = \sum_{m=1}^M N_t^m$. (N_t) is a Poisson process with intensity $\lambda = \sum_{m=1}^M \lambda_m$.
- Let $Q_t = \sum_{m=1}^M y_m N_t^m$. (Q_t) is a compound Poisson process equal in distribution to $\sum_{i=1}^{N_t} Y_i$, where (Y_i) is an i.i.d. sequence of random variables such that $\mathbf{P}(Y_i = y_m) = \frac{\lambda_m}{\lambda}$ for all $m = 1, \ldots, M$.
- Let $\beta = \sum_{m=1}^M \frac{\lambda_m}{\lambda} y_m = \mathbf{E}[Y_1]$. Then $Q_t - \lambda\beta t$ is a martingale.

Adding compound Poisson jumps to the Black & Scholes model II

- We consider the following extension of the Black & Scholes model:

$$\begin{aligned} dS_t &= \alpha S_t dt + \sigma S_t dB_t + S_{t-} d(Q_t - \lambda\beta t) \\ &= (\alpha - \lambda\beta) S_t dt + \sigma S_t dB_t + S_{t-} dQ_t. \end{aligned}$$

Proposition 7.12 (Asset price with jumps)

A solution to the above SDE is $S_t = S_0\, e^{\left(\alpha - \beta\lambda - \frac{\sigma^2}{2}\right)t + \sigma B_t} \prod_{i=1}^{N_t} (1 + Y_i)$.

- At the time t_i of the i-th jump, $S_{t_i} = (1 + Y_i) S_{t_i -}$.

Adding compound Poisson jumps to the Black & Scholes model III

- Let $\theta \in \mathbb{R}$ and $(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_M) \in (\mathbb{R}_+^*)^M$. Let

$$Z_0(t) = \exp\left(-\theta B_t - \frac{1}{2}\theta^2 t\right), \quad Z_m(t) = e^{(\lambda_m - \tilde{\lambda}_m)t}\left(\frac{\tilde{\lambda}_m}{\lambda_m}\right)^{N_t^m}, (m = 1, \ldots, M),$$

  and

$$Z(t) = Z_0(t)\prod_{m=1}^{M} Z_m(t).$$

- Z is a $\mathcal{F}_t$-martingale with respect to the measure $\mathbf{P}$, with expectation 1
- Using the Girsanov theorem and the above results on change of measure for Poisson processes, we may verify that under the measure $\mathbf{Q}$ defined by $\frac{d\mathbf{Q}}{d\mathbf{P}} = Z_T$,
  - $B^{\mathbf{Q}}$ defined by $B_t^{\mathbf{Q}} = B_t + \theta t$ is a $\mathcal{F}_t$-Brownian motion ;
  - $N_t^1, \ldots, N_t^M$ are M independent time-homogeneous Poisson processes with intensity $\tilde{\lambda}_m$;

Adding compound Poisson jumps to the Black & Scholes model IV

- Q is a compound Poisson process intensities $\tilde{\lambda} = \sum_{i=1}^{M} \tilde{\lambda}^m$ and distribution of jump sizes $\left(\frac{\tilde{\lambda}_m}{\tilde{\lambda}}\right)_{m=1,\dots,M}$;
- $B^{\mathbf{Q}}$ and Q are independent.

Proposition 7.13 (Equivalent martingale measure)

In the Black & Scholes extension with compound Poisson jumps with finitely-many jump sizes, the measure $\mathbf{Q}$ *defined above is an equivalent martingale measure if*

$$\alpha - r = \sigma\theta + \sum_{m=1}^{M} (\lambda_m - \tilde{\lambda}_m) y_m.$$

- Model completeness ?

Adding compound Poisson jumps to the Black & Scholes model V

- Let $\mathbf{Q}$ be an equivalent martingale measure and let $\tilde{\beta} = \mathbf{E}^{\mathbf{Q}}[Y_1]$.

Proposition 7.14 (Call price with compound Poisson jumps)

The price at time t of the standard call option with strike K and horizon T in this model is

$$C(t, S_t) = \sum_{j=0}^{\infty} e^{-\tilde{\lambda}(T-t)} \frac{\tilde{\lambda}^j (T-t)^j}{j!} \mathbf{E}^{\mathbf{Q}} \left[C_{BS} \left(S_t e^{-\tilde{\lambda}\tilde{\beta}(T-t)} \prod_{i=1}^{j} (1 + Y_i), \sigma, K, T - t \right) \right]$$

where $C_{BS}(s, \sigma, K, T)$ is the standard Black & Scholes formula.

- When using the delta hedge $\Delta_t = C_x(t, x)$, the hedging portfolio outperforms the option between jumps and hedges "in average".

Poisson jumps with log-normal sizes

- The previous methodology remains valid with continuously distributed jump sizes.
- Merton's original work was developed with log-normal jumps (Merton, 1976). If we assume log-normal jumps, i.e. $\log(1+Y_i)$ is Gaussian with mean μ_J and variance σ_J^2, since

$$S_t = S_0\, e^{\left(r-\tilde{\beta}\tilde{\lambda}-\frac{\sigma^2}{2}\right)t+\sigma B_t} \prod_{i=1}^{N_t}(1+Y_i) = S_0\, e^{\left(r-\tilde{\beta}\tilde{\lambda}-\frac{\sigma^2}{2}\right)t+\sigma B_t} e^{\sum_{i=1}^{N_t}\log(1+Y_i)},$$

  then the r.v. $\log \frac{S_t}{S_0} | N_t = j$ is Gaussian with mean $(r-\tilde{\beta}\tilde{\lambda}-\frac{\sigma^2}{2})t + j\mu_J$ and variance $\sigma^2 t + j\sigma_J^2$.
- In this special case, the call price formula above is thus further simplified to a series of Black & Scholes-type formulas.

TP Question VI

On considère un modèle à sauts log-normaux : les v.a. $\log(1 + Y_i)$ sont gaussiennes de moyenne μ_J et variance σ_J^2. On pourra prendre pour valeurs numériques par défaut $S0 = 100$, $r = 0.05$, $\sigma = 0.20$, $K = 100$, $T = 2$, $\tilde{\lambda} = 2$, $\mu_J = -0.02$ et $\sigma_J = 0.2$.

1. En terminant le raisonnement du slide précédent, écrire la formule de pricing explicite du prix du call dans le modèle avec sauts.
2. Etudier la convergence du prix en fonction du nombre de termes de la somme.
3. Implémenter le pricing du prix du call dans le modèle avec sauts et vérifier sa convergence vers le prix exact.
4. Tracer le smile de volatilité implicite engendré par le modèle à sauts et analyser. Commenter.

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Adding jumps to the stochastic volatility models

Stochastic volatility with jumps

- Practitionners often report the poor fit provided by the Heston model for short horizons (fitted curve is too flat).
- Mixing stochastic volatility and log-normal Poisson jumps, we may define the stochastic volatility with jumps (SVJ) model:

$$\begin{cases} dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dZ_1(t) + (e^{\alpha+\delta Z} - 1)dN_t, \\ dv_t = -\lambda(v_t - \bar{v})\,dt + \eta\sqrt{v_t}\,dZ_2(t), \end{cases}$$

  with Z_1 and Z_2 univariate Brownian motions such that $\langle Z_1, Z_2\rangle_t = \rho t$.
- Fourier/Characteristic function pricing is available for vanilla products, and Monte Carlo can be used for more complex products.

References

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